

Light-Load and Cross-Load Control of Integrated OBC and APM for Electric Vehicle Applications

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Abstract – To achieve high power density, onboard charger (OBC) and the auxiliary power module (APM) are electrically integrated in electric vehicles (EVs). A key challenge of the integration is the light-load and cross-load operation in simultaneous power transfer modes, for example, in the charging mode. Specifically in [1], at light loads, the OBC operates at very high switching frequency and dc bus-side switches tend to lose zero-voltage switching (ZVS). Moreover, the operation at very high switching frequency limits the maximum power which can be delivered to the low voltage (LV) output. In this paper, a control for light-load and cross-load operation in the charging mode for integrated OBC and APM topology, proposed in [1], is presented. A smooth transition from variable frequency (VF) to fixed frequency (FF) control and vice versa is achieved with power hysteresis control. The detailed block-diagram of control is provided. Finally, the experimental waveforms at light-loads, cross-loads and during transition from VF to FF control and vice versa in charging mode of operation are provided. At light load operation of OBC, at $V_{Bus} = 400V$, $V_{HV} = 370V$, $V_{LV} = 13.8V$, $P_{LV} = 240W$, measured efficiencies of 96.4% and 94.4% are achieved at $P_{HV} = 2.67 kW$ and $P_{HV} \approx 550 W$, respectively.

I. INTRODUCTION

Recently, a lot of research is being published about achieving high power density power conversion in electric vehicles (EVs) by integrating the onboard charger (OBC) and the auxiliary power module (APM) [1]-[4]. The integration of OBC and APM helps to achieve high power density by reducing the overall number of switching and magnetic components. However, a key challenge of integration is light load control in simultaneous power transfer modes, for example, during charging mode when high voltage (HV) and low voltage (LV) batteries are simultaneously charged from the grid. Also, in the simultaneous power transfer modes, the proper operation must be ensured in cross-load conditions, i.e., when load at one port is heavy and the load at the other port is very light or no load. In literature, the detailed operation at light-loads or cross-loads in simultaneous power transfer modes has not been shown. Specifically, in the topology for integration of 6.6kW OBC and 2.5kW APM proposed in [1], shown in Fig. 1, the design of control for light-load and cross-load operation is very important. In the charging mode, when the load at OBC output is heavy, OBC operates as a series resonant converter (SRC) with variable switching frequency and delay-time control and the power is

transferred to the LV battery with a dual active bridge (DAB) operation, i.e., by phase shifting LV-side switches with respect to dc bus-side switches. When the load at OBC output becomes lighter, the switching frequency increases above the maximum switching frequency which is required to deliver maximum power to the LV battery. Therefore, at light load of OBC, maximum power cannot be delivered to the LV output. Also, at light loads, the dc bus-side switches lose zero voltage switching (ZVS) which, at high switching frequencies, causes very high switching losses that result in heating up and eventual damage of the devices.

To perform light load control of a power converter, one of the well-known techniques is the burst-mode control. However, applying burst mode control for integrated OBC and APM in simultaneous power transfer modes will reduce the load at both the outputs and independent regulation during cross-loads cannot be achieved. Moreover, operating with burst mode results in high ripple of output voltages, poor EMI, and audible noise [5]-[7].

In this paper, a control for light-load and cross-load operation in simultaneous power transfer mode for integrated OBC and APM topology, proposed in [1], is presented. When the load at the HV output becomes lighter, the switching frequency is reduced and kept fixed. The voltage at the HV output is regulated by controlling the shorting time of two HV-side switches, whereas, the voltage at the LV output is regulated by phase-shifting LV-side switches with respect to the dc-bus side switches. A smooth transition from variable-frequency (VF) to fixed-frequency (FF) control and vice versa is achieved with power hysteresis control. The detailed block-diagram of control is provided and the control is well explained with the help of Simplis simulation waveforms. Finally, the experimental waveforms at light-loads, cross-loads, and during the transition from VF to FF control and vice versa in charging mode are provided. At light load operation of OBC, at $V_{Bus} = 400V$, $V_{HV} = 370V$, $V_{LV} = 13.8V$, $P_{LV} = 240W$, measured efficiencies of 96.4% and 94.4% are achieved at $P_{HV} = 2.67 kW$ and $P_{HV} \approx 550 W$, respectively.

II. PROPOSED CONTROL FOR LIGHT-LOAD AND CROSS-LOAD OPERATION

Figure 2 shows the control plot of integrated OBC and APM, proposed in [1], in the charging mode of operation. The plot shows LV output power P_{LV} vs. HV output power P_{HV} . It can be seen in Fig. 2 that when P_{HV} is greater than a threshold power level P_{TH} , HV battery voltage is controlled by variable frequency and delay-time control and LV battery is controlled by the phase-shift control as explained before, whereas, when P_{HV} is smaller than P_{TH} , light-load control

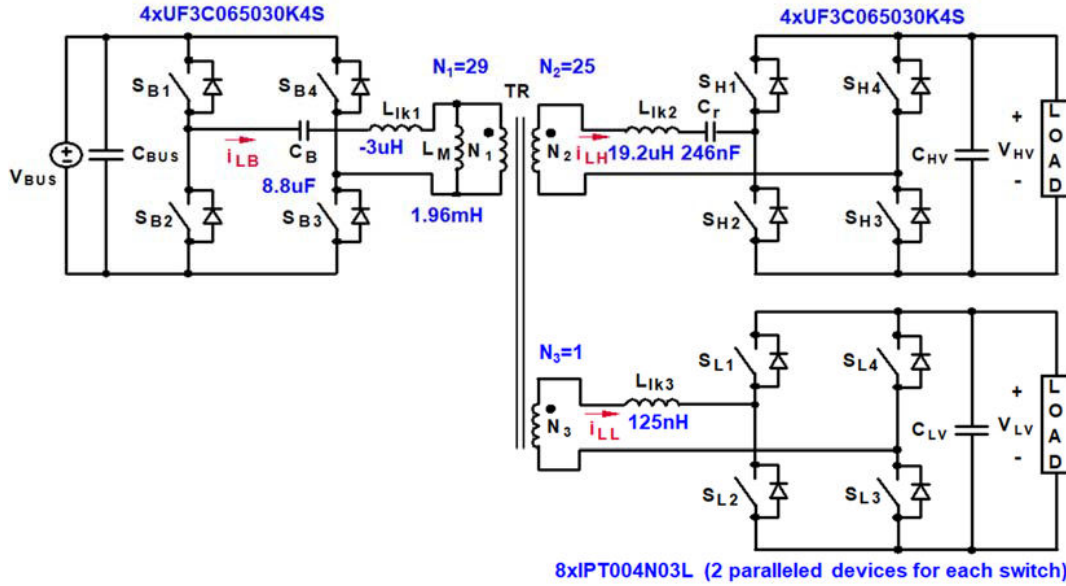


Fig. 1. Experimental circuit diagram of integrated 6.6 kW OBC and 2.5 kW APM proposed in [1].

must be performed. It should be noted that the value of P_{TH} is determined based on P_{HV} at which the switching frequency of operation is smaller than the maximum switching frequency $f_{swmax}=170$ kHz which is required to deliver maximum power of 1kW to the LV output. Also, it must be ensured that at P_{TH} , the dc bus-side switches still can achieve or just start to lose ZVS when P_{LV} is very small ($<20\%$ of the maximum P_{LV}). It should be noted that dc-bus side switches achieve ZVS based on both P_{HV} and P_{LV} . In order to simplify control, P_{TH} is determined based on P_{HV} at which the dc-bus side switches just start to lose ZVS when P_{LV} is very small. In this case, when P_{LV} is close to its maximum value, dc-bus side switches may achieve ZVS but still the control will be transitioned to the light load. Figure 3 shows the key experimental waveforms in charging mode at $V_{BUS}=400$ V, $V_{HV}=370$ V, $P_{HV}=2.7$ kW, $V_{LV}=13.8$ V, $P_{LV}\approx 180$ W. It can be seen that the switching frequency $f_{sw}=158$ kHz which is just smaller than f_{swmax} . As seen in Fig. 3, the drain-source voltages of dc bus-side switches S_{B3} and S_{B2} , V_{DSB3} and V_{DSB2} , respectively, operate with 50% duty cycle, variable switching frequency pulses and the drain-source voltage of switch S_{H3} , V_{DSH3} is delayed by delay time T_{delay} with respect to drain-source voltage of switch S_{H2} , V_{DSH2} . During T_{delay} ,

the HV full-bridge terminals are shorted and the energy is stored at the resonant inductor. At the end of the delay-time, when switch S_{H3} turns on, the energy stored at the resonant inductor is delivered to the output. The drain-source voltages of switches S_{L3} and S_{L2} , V_{DSL3} and V_{DSL2} operate at the same switching frequency but are phase-shifted with respect to V_{DSB3} and V_{DSB2} , respectively to regulate V_{LV} . It can be seen from the waveforms of V_{DSB3} and V_{DSB2} in Fig. 3, at this power level, dc bus-side switches just start to lose ZVS. This is because at the switch turn off/turn on instant, the dc bus-side current I_{LB} is small to fully charge/discharge the output capacitances of the corresponding switches. Therefore, when P_{HV} becomes smaller than $P_{TH}=2.7$ kW, the control must be transitioned to the light load control. It should be noted that the value of P_{TH} will be different at different values of V_{HV} and must be determined either through simulations or experiments.

The proposed light-load control is described with the help of key experimental waveforms in charging mode at $V_{BUS}=400$ V, $V_{HV}=370$ V, $P_{HV}=2.7$ kW, $V_{LV}=13.8$ V, and $P_{LV}\approx 240$ W in Figure 4. As seen in Fig. 4, at light load, the

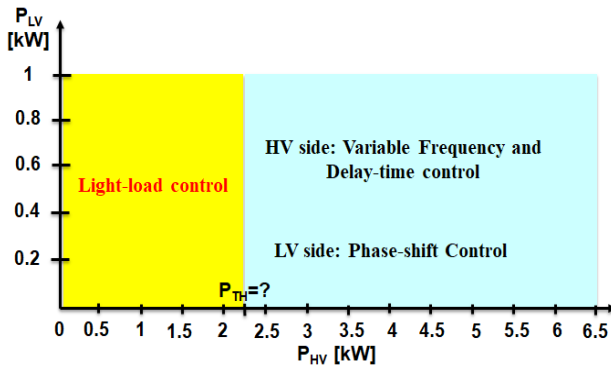


Fig. 2. Control plot of integrated OBC and APM, proposed in [1], in charging mode of operation.

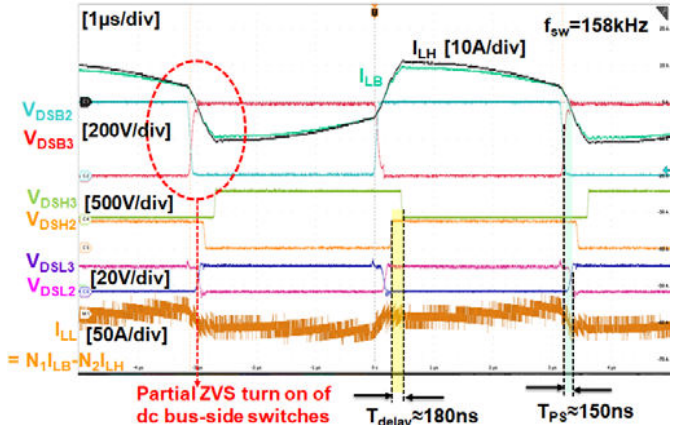


Fig. 3. Key experimental waveforms in charging mode at $V_{BUS}=400$ V, $V_{HV}=370$ V, $P_{HV}=2.7$ kW, $V_{LV}=13.8$ V, $P_{LV}=180$ W.

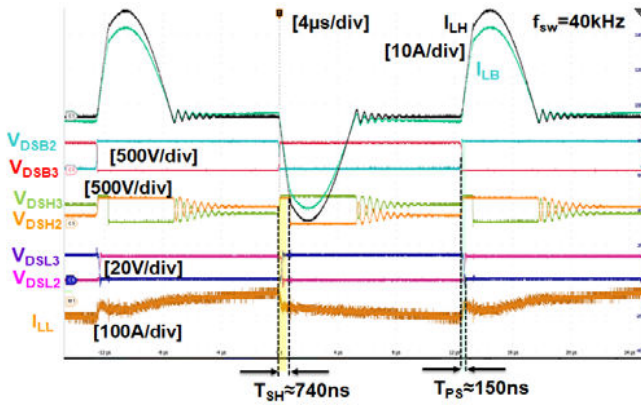


Fig. 4. Key experimental waveforms in charging mode at $V_{BUS} = 400$ V, $V_{HV} = 370$ V, $P_{HV} = 2.7$ kW, $V_{LV} = 13.8$ V, $P_{LV} = 240$ W with proposed light-load control.

switching frequency is kept constant at a reduced value of 40 kHz, which is selected based on the transformer design. Voltages V_{DSB2} and V_{DSB3} operate with 50% duty-cycle pulses. In order to regulate the HV output voltage, the HV full-bridge terminals are shorted for a shorting time T_{SH} , by making V_{DSH2} and V_{DSH3} high together. Voltages V_{DSH2} and V_{DSH3} are controlled by controlling switches S_{H1} and S_{H4} , respectively with fixed switching frequency, 180-degree phase-shifted pulses with on-time equal to T_{SH} . It should be noted that switches S_{H2} and S_{H3} are disabled so that they only operate as rectifiers. During the shorting time T_{SH} , the energy is stored at the resonant tank. At the end of the highlighted shorting time in Fig. 4, when V_{DSH2} becomes zero, i.e., when switch S_{H1} turns off, the stored energy at the resonant tank is delivered to the output. During this time, the resonant inductor current I_{LH} decreases. Finally, when I_{LH} becomes zero, since all the switches on the HV-side are off, the body diodes of the switches stop conducting and the resonant inductor current remains zero until the next switching cycle. V_{LV} is regulated by phase-shifting V_{DSL3} and V_{DSL2} with respect to V_{DSB3} and V_{DSB2} , respectively by time T_{PS} . Since, HV and LV outputs are controlled with two different control variables T_{SH} and T_{PS} , respectively, independent control of both output ports can be achieved at all light-load and cross-load conditions. It should also be noted that with the proposed light load control in Fig. 4, all the dc-bus and HV-side switches lose ZVS and therefore, the switching frequency is reduced at light load to a minimum value so as to keep switching losses low and prevent heating up of the devices. Also, because the power is still delivered to the output every switching cycle, the output voltage ripple at the HV-output is much smaller as compared to that with the conventional burst mode operation. Figures 5 (a), 5(b), and 5(c) show key experimental waveforms in charging mode at $V_{BUS} = 400$ V, $V_{HV} = 370$ V, $V_{LV} = 13.8$ V, at cross-load conditions $P_{HV} = 500$ W and $P_{LV} \approx 540$ W, $P_{HV} = 2.3$ kW and $P_{LV} \approx 540$ W, $P_{HV} = 2.3$ kW and $P_{LV} = 66$ W, and $P_{HV} = 500$ W and $P_{LV} = 66$ W, respectively. The regulation of both V_{HV} and V_{LV} is achieved at all the cross-load conditions by controlling time T_{SH} and T_{PS} independently and keeping switching frequency fixed at

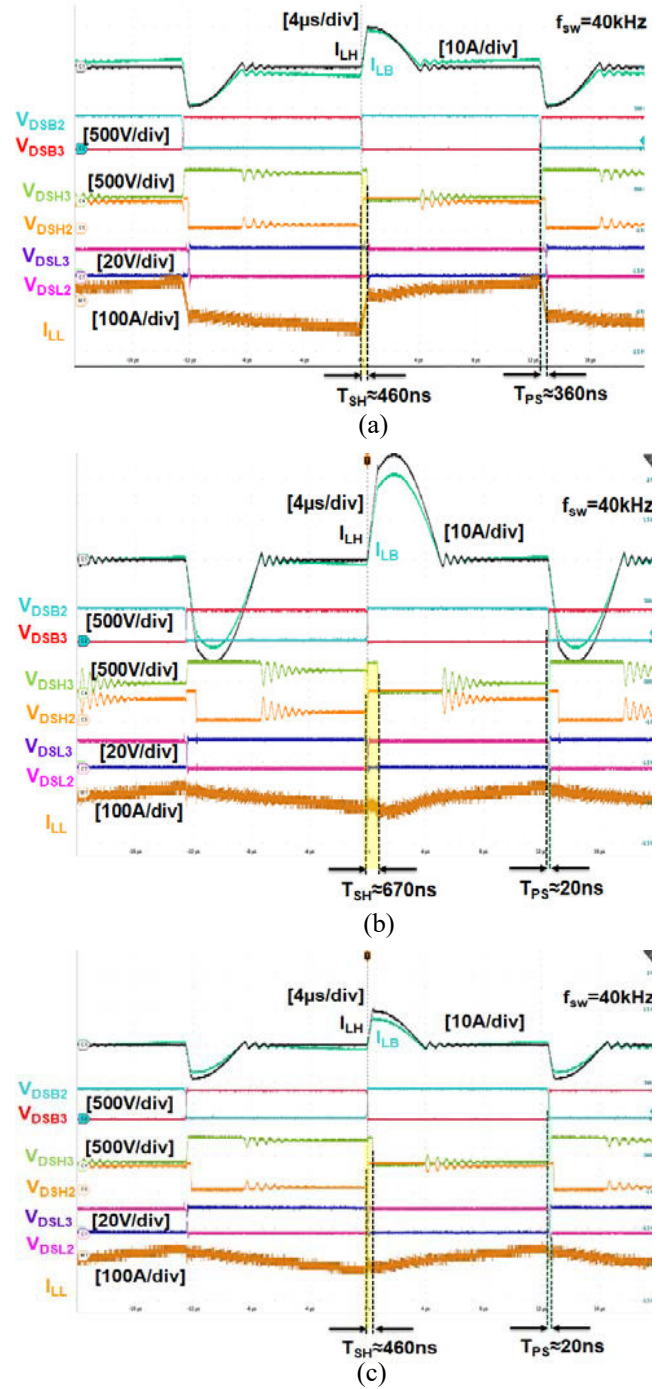


Fig. 5. Key experimental waveforms in charging mode at $V_{BUS} = 400$ V, $V_{HV} = 370$ V, $V_{LV} = 13.8$ V at cross-load conditions a) $P_{HV} = 500$ W, $P_{LV} = 540$ W, b) $P_{HV} = 2.3$ kW, $P_{LV} = 66$ W, and c) $P_{HV} = 500$ W, $P_{LV} = 66$ W.

$f_{sw} = 40$ kHz. The operation in Fig. 5 at different cross-load conditions is achieved in similar way as explained in Fig. 4.

Figure 6 shows the detailed block diagram of control of integrated OBC and APM in charging mode of operation. The HV output is regulated by a voltage controller. The output of the voltage controller $V_{EA(HV)}$ controls the switching frequency f_{sw} when $V_{EA(HV)}$ is larger than a threshold value V_{EATH} . This region is called the variable frequency (VF) control region. When $V_{EA(HV)}$ is smaller than

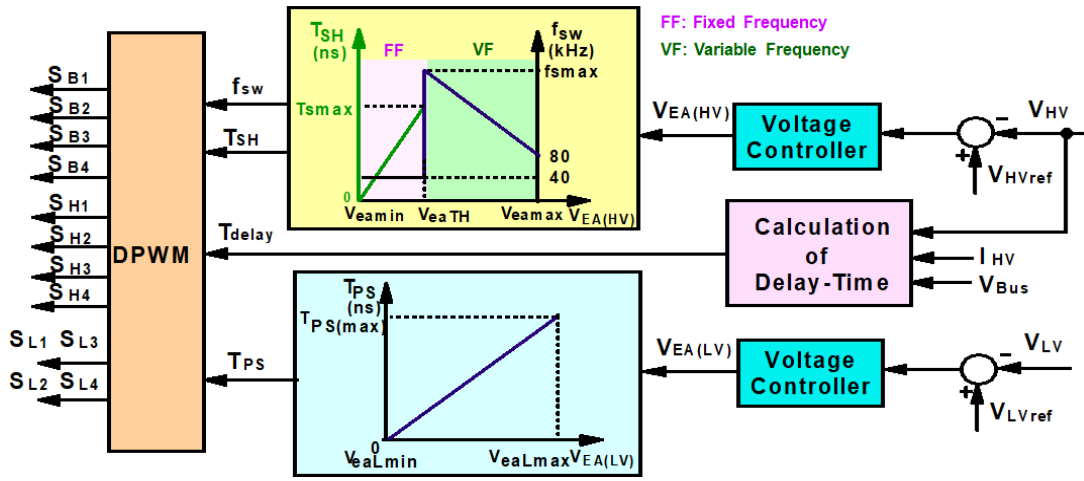


Fig. 6. Block diagram of control of integrated OBC and APM in charging mode of operation.

V_{EATH} , $V_{EA(HV)}$ controls the shorting time T_{SH} of the HV-side switches while f_{sw} is fixed. This region is called the fixed frequency (FF) control region. It should be noted that at a particular HV voltage, $V_{EA(HV)}$ directly controls P_{HV} and therefore, the threshold V_{EATH} is proportional to P_{TH} . The LV output is regulated by another voltage controller. The output of the voltage controller $V_{EA(LV)}$ controls the phase-shift T_{PS} of LV-side switches with respect to the dc bus-side switches. The digital pulse width modulator (DPWM) of the DSP generates the switching pulses of the dc bus-side switches based on f_{sw} , the switching pulses of HV-side switches based on f_{sw} and T_{delay} in VF control region and based on f_{sw} and T_{SH} in FF control region, and the switching pulses of LV-side switches based on f_{sw} and T_{PS} . It should be noted that the delay-time T_{delay} is calculated in VF control region from values of V_{Bus} , V_{HV} , and output current I_{HV} , based on the calculations in [8] and [9] and then implemented with help of a look-up table. It should also be noted that at the transition of control from VF to FF mode and vice versa, the maximum shorting-time T_{smax} in the FF mode must be determined to deliver the same output power which corresponds to the maximum switching frequency f_{smax} in the VF mode. For example, from Fig. 4, the required value of $T_{smax} = 740$ ns in FF mode at $P_{TH} = 2.7$ kW. At the same output power, $P_{TH} = 2.7$ kW, the maximum switching frequency in VF mode is $f_{smax} = 158$ kHz as seen in Fig. 3. If the value of T_{smax} is even slightly lower than its required value in FF mode, the control jitters between the two modes. It should be noted that the value of T_{smax} can be determined either experimentally or through simulations.

Figure 7 shows key simulation waveforms at $V_{Bus} = 400$ V, $V_{HV} = 370$ V, $P_{HV} = 2.7$ kW, $V_{LV} = 13.8$ V, and $P_{LV} = 250$ W with only 3 ns smaller value of T_{smax} than its required value. At time t_0 , the control is operating in VF mode, therefore, when $V_{HV} > V_{HVref}$, $V_{EA(HV)}$ decreases and f_{sw} increases. At time t_1 , $V_{EA(HV)} < V_{EATH}$, the control transitions to FF mode. Because, at this instant, T_{smax} is smaller than its required value, the power delivered to the output is smaller than that required by the load. Therefore, V_{HV} decreases. As V_{HV} decreases below V_{HVref} , at time t_2 , $V_{EA(HV)}$ starts

increasing. Finally, at time t_3 , when $V_{EA(HV)} > V_{EATH}$, the control transitions again to VF mode. In this way, the control jitters between the VF and FF mode. This jitter between two modes increases the peak-peak HV and LV output voltage ripples to ~ 1.4 V and ~ 0.6 V, respectively. In order to have a smooth transition between the two control modes, the power hysteresis is implemented.

III. SMOOTH TRANSITION BETWEEN VARIABLE FREQUENCY AND FIXED FREQUENCY CONTROL

To implement power hysteresis, a slightly larger than the required value of T_{smax} in FF mode at $P_{HV} = 2.7$ kW is applied. Figure 8 shows key simulation waveforms when control transitions from VF to FF mode at $V_{Bus} = 400$ V, $V_{HV} = 370$ V, $P_{HV} = 2.7$ kW, $V_{LV} = 13.8$ V, and $P_{LV} = 250$ W in charging mode with 25 ns larger value of T_{smax} than its required value. At time t_0 , the control is operating in VF mode, therefore, when $V_{HV} > V_{HVref}$, $V_{EA(HV)}$ decreases. At time t_1 , when $V_{EA(HV)} < V_{EATH}$, the control mode transitions from VF to FF mode. In FF mode, since value of T_{smax} is larger than required value, the power delivered to the output is larger and as a result V_{HV} increases. Because $V_{HV} > V_{HVref}$, $V_{EA(HV)}$ further decreases below V_{EATH} and settles at a new

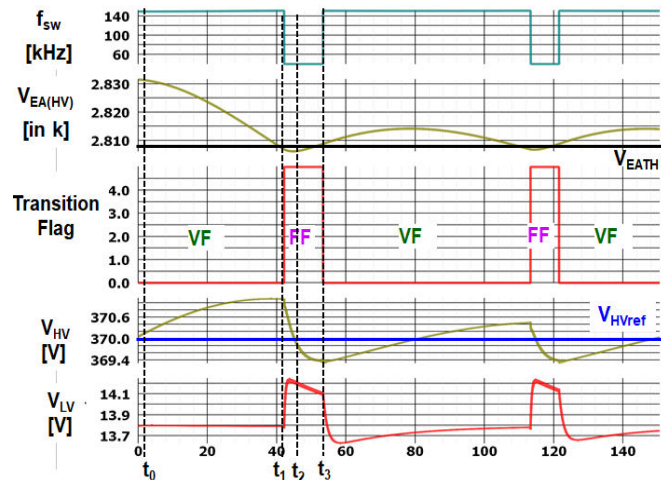


Fig. 7. Key simulation waveforms at $V_{Bus} = 400$ V, $V_{HV} = 370$ V, $P_{HV} = 2.7$ kW, $V_{LV} = 13.8$ V, $P_{LV} = 250$ W with only 3 ns smaller value of T_{smax} than its required value.

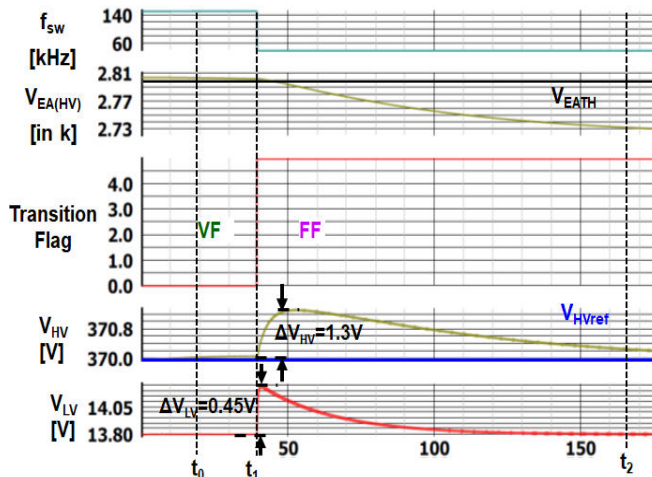


Fig. 8. Key simulation waveforms when control transitions from VF to FF mode at $V_{BUS} = 400$ V, $V_{HV} = 370$ V, $P_{HV} = 2.7$ kW, $V_{LV} = 13.8$ V, $P_{LV} = 250$ W with 25 ns larger value of T_{smax} than its required value.

steady state value at time t_2 . In this way, the control smoothly transitions from VF to FF mode without any jitter. At the transition of control, the overshoot in V_{HV} , $\Delta V_{HV} = 1.3$ V and the overshoot in V_{LV} , $\Delta V_{LV} = 0.45$ V.

Figure 9 shows key simulation waveforms when control transitions from FF to VF mode at $V_{BUS} = 400$ V, $V_{HV} = 370$ V, $P_{HV} = 3.1$ kW, $V_{LV} = 13.8$ V, and $P_{LV} = 250$ W in charging mode with 25ns larger value of T_{smax} than its required value. At time t_0 , the control is operating in FF mode, therefore, when $V_{HV} < V_{HVref}$, $V_{EA(HV)}$ increases. At time t_1 , when $V_{EA(HV)} > V_{EATH}$, the control mode transitions from FF to VF mode. In VF mode, at time t_1 , since the maximum switching frequency f_{smax} delivers $P_{HV} = 2.7$ kW but the output power demanded by the load is $P_{HV} = 3.1$ kW, therefore, the HV voltage V_{HV} decreases. Because $V_{HV} < V_{HVref}$, $V_{EA(HV)}$ further increases above V_{EATH} and settles at a new steady state value at time t_2 . In this way, the control smoothly transitions from FF to VF mode without any jitter. At the transition of control, undershoot in V_{HV} , $\Delta V_{HV} = 2.5$ V and undershoot in V_{LV} , $\Delta V_{LV} = 0.45$ V. It should

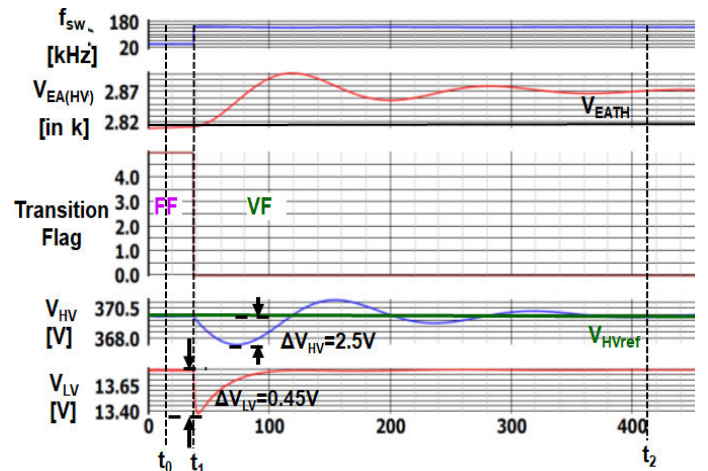


Fig. 9. Key simulation waveforms when control transitions from FF to VF mode at $V_{BUS} = 400$ V, $V_{HV} = 370$ V, $P_{HV} = 3.1$ kW, $V_{LV} = 13.8$ V, $P_{LV} = 250$ W with 25 ns larger value of T_{smax} than its required value.

be noted that because T_{smax} in FF mode is larger than its required value at $P_{HV} = 2.7$ kW, when the control starts from FF mode, it operates in FF mode till $P_{HV} = 3.1$ kW and then transitions to the VF mode. In this way, power hysteresis band is ~ 400 W in the transition of control. It should be also noted that the power hysteresis band can be changed by changing the value by which T_{smax} is increased over its required value.

By performing non-linear feedforward control at the transition from VF to FF mode and from FF to VF mode, overshoots and undershoots in HV voltage can be significantly reduced. The non-linear feedforward control is implemented by multiplying output of voltage controller $V_{EA(HV)}$ with a constant K at the transition of the control. The value of constant K is the ratio of the new steady state value of $V_{EA(HV)}$ after the control transitions to the threshold value V_{EATH} . Figure 10 shows key simulation waveforms when control transitions from VF to FF mode at $V_{BUS} = 400$ V, $V_{HV} = 370$ V, $P_{HV} = 2.7$ kW, $V_{LV} = 13.8$ V, and $P_{LV} = 250$ W in charging mode with 25ns larger value of T_{smax} than its required value and constant $K = 0.972$ multiplied with voltage controller output at the transition of control. As seen in Fig. 10, by directly multiplying the output of the voltage controller with $K = 0.972$ at the instant when control transitions from VF to FF mode, the output of voltage controller $V_{EA(HV)}$ is close to its steady state value at the transition instant. As a result, the undershoot in V_{HV} , $\Delta V_{HV} = 0.2$ V which is significantly reduced as compared to $\Delta V_{HV} = 1.3$ V in Fig. 8. Figure 11 shows key simulation waveforms when control transitions from FF to VF mode at $V_{BUS} = 400$ V, $V_{HV} = 370$ V, $P_{HV} = 3.1$ kW, $V_{LV} = 13.8$ V, and $P_{LV} = 250$ W in charging mode with 25ns larger value of T_{smax} than its required value and constant $K = 1.025$ multiplied with voltage controller output at the transition of control. As seen in Fig. 11, by directly multiplying the output of the voltage controller with $K = 1.025$ at the instant when control transitions from FF to VF mode, the output of voltage controller $V_{EA(HV)}$ is close to its steady state value at the transition instant. As a result, the overshoot in V_{HV} ,

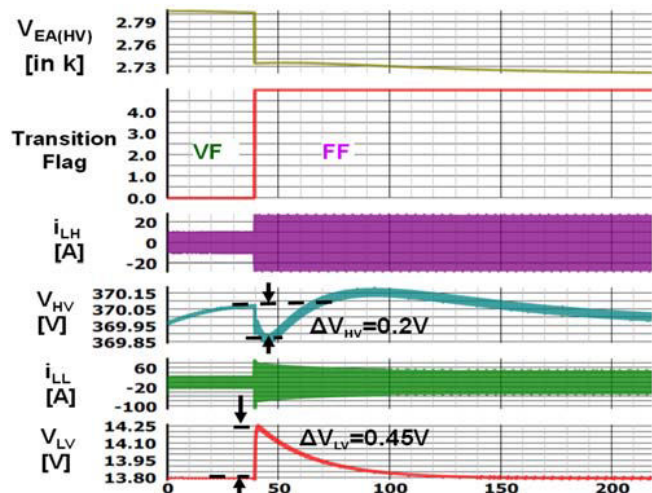


Fig. 10. Key simulation waveforms when control transitions from VF to FF mode at $V_{BUS} = 400$ V, $V_{HV} = 370$ V, $P_{HV} = 2.7$ kW, $V_{LV} = 13.8$ V, $P_{LV} = 250$ W with 25 ns larger value of T_{smax} than its required value and constant $K = 0.972$ multiplied with voltage controller output at transition of control.

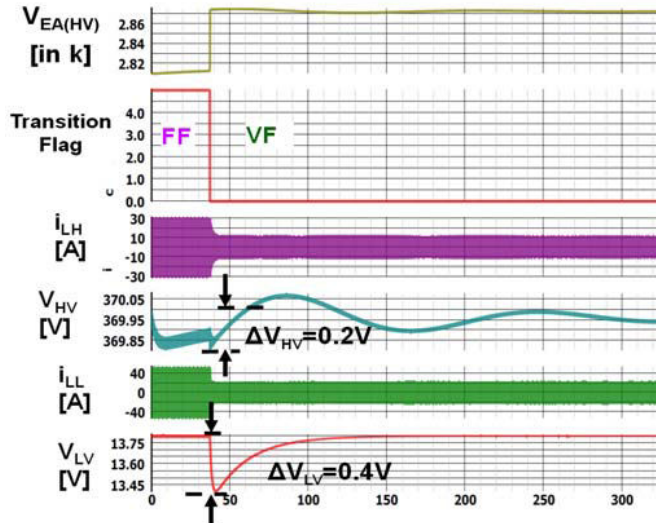


Fig. 11. Key simulation waveforms when control transitions from FF to VF mode at $V_{BUS} = 400$ V, $V_{HV} = 370$ V, $P_{HV} = 3.1$ kW, $V_{LV} = 13.8$ V, $P_{LV} = 250$ W with 25 ns larger value of T_{smax} than its required value and constant $K=1.025$ multiplied with voltage controller output at transition of control.

$\Delta V_{HV} = 0.2$ V which is significantly reduced as compared to $\Delta V_{HV} = 2.5$ V in Fig. 9.

IV. EXPERIMENTAL RESULTS

Figure 12 shows key experimental waveforms when control transition from VF to FF mode at $V_{BUS} = 400$ V, $V_{HV} = 370$ V, $P_{HV} = 2.8$ kW, $V_{LV} = 13.6$ V, and $P_{LV} = 574$ W in charging mode with 30ns larger value of T_{smax} than its required value. It can be seen that the control transitions smoothly from VF to FF mode without any jitter. Figure 13 shows key experimental waveforms when control transition from FF to VF mode at $V_{BUS} = 400$ V, $V_{HV} = 370$ V, $P_{HV} = 3.067$ kW, $V_{LV} = 13.6$ V, and $P_{LV} = 574$ W in charging mode with 30ns larger value of T_{smax} than its required value. It can be seen that the control transitions smoothly from FF to VF mode without any jitter. It can be observed in both Figs. 12 and 13 that the control transitions from VF to FF mode and back from FF to VF mode within one switching cycle. In experiments, with 30ns larger value of T_{smax} than its required value at the transition instant, the power hysteresis band is ~ 267 W.

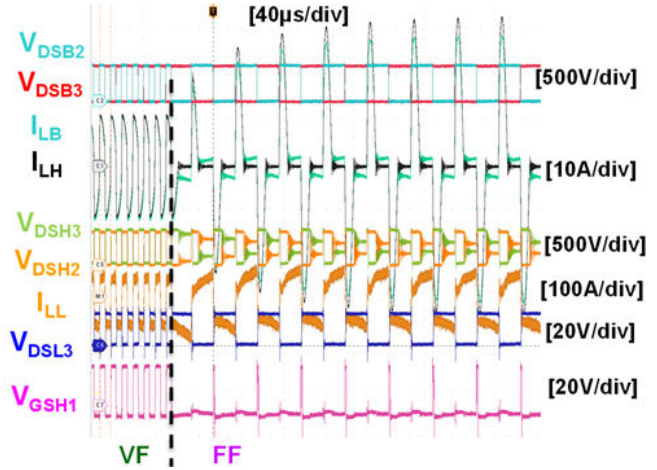


Fig. 12. Key experimental waveforms when control transitions from VF to FF mode at $V_{BUS} = 400$ V, $V_{HV} = 370$ V, $P_{HV} = 2.8$ kW, $V_{LV} = 13.6$ V, $P_{LV} = 574$ W.

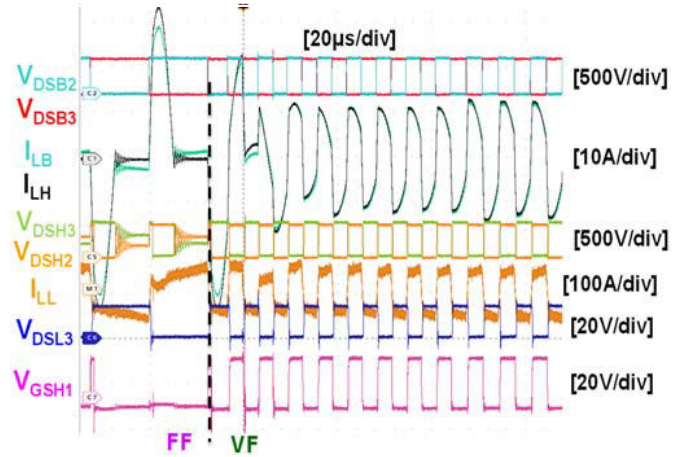


Fig. 13. Key experimental waveforms when control transitions from FF to VF mode at $V_{BUS} = 400$ V, $V_{HV} = 370$ V, $P_{HV} = 3.067$ kW, $V_{LV} = 13.6$ V, $P_{LV} = 574$ W.

Figures 14(a) and 14(b) show key experimental waveforms at $V_{BUS} = 400$ V, $V_{HV} = 370$ V, $V_{LV} = 13.6$ V, and $P_{LV} = 574$ W in charging mode at $P_{HV} = 2.8$ kW during control transition from VF to FF mode and at $P_{HV} = 3.067$ kW during control transition from FF to VF mode, respectively. In Fig. 14(a), the undershoot in V_{HV} , $\Delta V_{HV} = 2$ V and the undershoot in V_{LV} , $\Delta V_{LV} = 0.3$ V during transition from VF to FF mode. In Fig. 14(b), the overshoot in V_{HV} , $\Delta V_{HV} = 1.5$ V and the

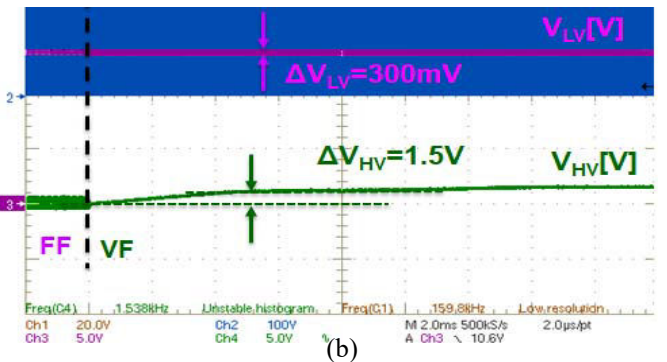
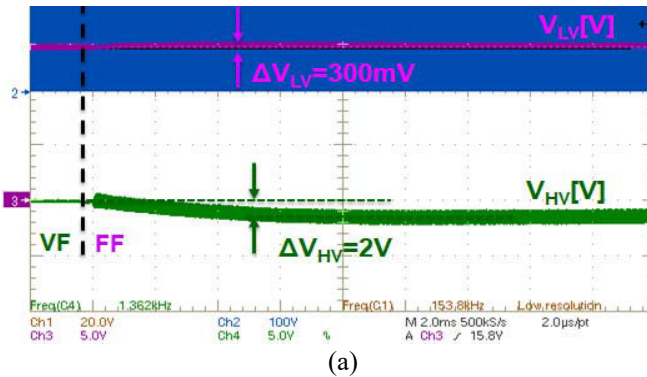


Fig. 14. Key experimental waveforms at $V_{BUS} = 400$ V, $V_{HV} = 370$ V, $V_{LV} = 13.6$ V, $P_{LV} = 574$ W in charging mode a) at $P_{HV} = 2.8$ kW during control transition from VF to FF mode and b) at $P_{HV} = 3.067$ kW during control transition from FF to VF mode.

overshoot in V_{LV} , $\Delta V_{LV} = 0.3V$ during transition from FF to VF mode. The overshoot/undershoot in V_{HV} is well below 1% of the steady-state voltage of 370V and the overshoot/undershoot in V_{LV} is approximately 2.2% of the steady-state voltage of 13.8V. It should be noted that these experimental waveforms are obtained with non-linear feedforward control at the transition of control by multiplying the output of the voltage controller with $K = 0.94$ at the instant when control transitions from VF to FF mode and by multiplying the output of the voltage controller with $K = 1.06$ at the instant when control transitions from FF to VF mode.

At light load operation of OBC, at $V_{Bus} = 400V$, $V_{HV} = 370V$, $V_{LV} = 13.8V$, $P_{LV} = 240W$, measured efficiencies of 96.4% and 94.4% are achieved at $P_{HV} = 2.67\text{ kW}$ and $P_{HV} \approx 550\text{ W}$, respectively.

V. CONCLUSION

In this paper, a control for light-load and cross-load operation in simultaneous power transfer mode for integrated OBC and APM, proposed in [1], is presented. When the load at the HV output becomes lighter, the switching frequency is reduced and kept fixed. The voltage at the HV output is regulated by controlling the shorting time of two HV-side switches, whereas, the voltage at the LV output is regulated by phase-shifting LV-side switches with respect to the dc-bus side switches. A smooth transition from variable-frequency (VF) to fixed-frequency (FF) control and vice versa is achieved with power hysteresis control. The detailed block-diagram of control is provided and the control is well explained with the help of Simplis simulation waveforms.

Finally, the experimental waveforms at light-loads, cross-loads, and during the transition from variable to fixed frequency control and vice versa are provided. At light load

operation of OBC, at $V_{Bus} = 400V$, $V_{HV} = 370V$, $V_{LV} = 13.8V$, $P_{LV} = 240W$, measured efficiencies of 96.4% and 94.4% are achieved at $P_{HV} = 2.67\text{ kW}$ and $P_{HV} \approx 550\text{ W}$, respectively.

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